THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Out of all rectangles with perimeter 32 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 80 **(b)** 64 **(c)** 72 **(d)** 56

2. A rod $3\frac{1}{2}$ m long has $\frac{3}{4}$ m cut off. The remaining length in metres is: 1

(a) $\frac{1}{4}$ **(b)** $4\frac{1}{4}$ **(c)** $2\frac{3}{4}$ **(d)** $2\frac{1}{4}$

3. An isosceles triangle OAB with OA = OB has $\angle AOB = 40^\circ$. Each base angle (degrees) is: 1

(a) 65 **(b)** 80 **(c)** 70 **(d)** 75

4. A regular polygon is rotated again and again through its smallest angle 40°. The number of turns needed to complete a full turn is: 1

(a) 8 **(b)** 9 **(c)** 11 **(d)** 10

5. If x + (-7) = -12, then x is: 1

(a) -7 **(b)** -5 **(c)** 5 **(d)** -3

6. The sides of a triangle are 13 cm, 14 cm and 15 cm. Its perimeter (cm) is: 1

(a) 55 **(b)** 42 **(c)** 43 **(d)** 41

7. The LCM of the denominators 6, 8 is used to add $\frac{1}{6}$ and $\frac{1}{8}$. The LCM is: 1

(a) 23 **(b)** 30 **(c)** 25 **(d)** 24

8. The diameter of the circle that exactly fits inside a square of side 10 cm is (cm): 1

(a) 11 **(b)** 10 **(c)** 15 **(d)** 9

9. A regular 6-gon has side 7 cm. Its perimeter (cm) is: 1

(a) 49 **(b)** 42 **(c)** 48 **(d)** 35

10. The value of 7 - 12 - (-7) is: 1

(a) 0 **(b)** 2 **(c)** 12 **(d)** 4

11. A square park of side 20 m has a footpath 1 m wide all around outside it. The area of the footpath (m^2) is: 1

(a) 84 **(b)** 92 **(c)** 80 **(d)** 88

12. $\frac{7}{9} + \frac{2}{3} - \frac{5}{18} =$ 1

(a) $\frac{4}{9}$ **(b)** $\frac{7}{18}$ **(c)** $1\frac{1}{6}$ **(d)** $1\frac{13}{18}$

13. Two circles have radii 13 cm and 5 cm and their centres are 8 cm apart. The circles: 1

(a) touch externally **(b)** touch internally

(c) do not touch (separate) **(d)** one lies inside the other

14. The image of a point 5 units from a mirror line is how far from the original point (units)? 1

(a) 5 **(b)** 11 **(c)** 10 **(d)** 15

15. The sum of the integers from -6 to 7 is: 1

(a) 6 **(b)** 13 **(c)** 7 **(d)** 8

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A carpet 5 m by 4 m is laid in a room 8 m by 6 m. Find the area of the uncovered floor. 2

17. A boy walks $\frac{3}{4}$ km to school and $\frac{2}{5}$ km to a shop, then returns by the same route. Find the total distance. 2

18. A square ABCD of side 10 cm has diagonals meeting at O. Find the area of triangle AOB and the sum of areas of the four triangles. 2

19. Reflect the point (2, 5) in 2

(a) the y-axis,

(b) the x-axis and give the coordinates.

20. A mountain climber starts at -20 m, goes up 60 m and then down 25 m. Find the climber's final height and its vertical distance from a camp at 30 m. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Write the next three terms of the pattern -20, -14, -8, -2, ... and find the first positive term. 3

22. A rectangular plot 50 m by 30 m is to be fenced with 2 rounds of barbed wire costing ₹15 per metre. Find the total cost. Also find the cost of ploughing it at ₹5 per m^2. 3

23. A school has 360 students. $\frac{4}{9}$ of them are girls. $\frac{3}{8}$ of the girls play chess. How many girls play chess and how many boys are there? 3

24. Four circles of radius 4 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

OR

A designer wants a symmetrical logo with 6 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 6 petals are placed. 3

25. A submarine at -10 m dives 70 m, rises 55 m, then dives 30 m. Find its position after each move and the total vertical distance moved. 3

OR

A rectangle 12 cm by 27 cm and a square have equal areas. Find the side of the square and compare their perimeters. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two regular polygons have 3 and 4 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). **5**

OR

A lift in a building with 3 basement floors starts at floor 4. It goes 6 floors down, 2 floors up, 3 floors down. State its final floor, and whether the final floor exists (the lowest basement is -3). **5**

27. A rectangle with length 24 cm and breadth 10 cm has a triangular piece with base 24 cm and height 6 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. **5**

OR

A trader sold $\frac{1}{4}$ of his cloth on day 1 and $\frac{2}{3}$ of the remaining on day 2. If 12 metres are left, find the total length of cloth. **5**

END OF QUESTION PAPER — Total: 50 Marks

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